Lab 5

1. Use the salary data
2. Handles Incomplete, noise and inconsistent values

Data Preprocessing-Cleaning

1. Incomplete (Missing Values)
2. Noise (Either errors)
3. Inconsistent

```
import numpy as np
import pandas as pd

#pd.read_csv() is a function called from pandas library
#df = pd.read_csv('salary.csv') # this command load salary.csv onto your dataframe named as df.

#You can mount the file on your google drive and load it. To make life easier, just make sure your directory doesn't have spaces
#Example    /Dataset/machine learning <<<This directory has spaces

from google.colab import drive
drive.mount('/content/drive')

df = pd.read_csv('/content/drive/MyDrive/Dataset/salary.csv') #load the csv dataset into a 'dataframe' df

#Convert "salary" string to integers
df['Salary'] = df['Salary'].replace(',','',regex=True) #this replace the comma ',' in 55,000 to 55000
df['Salary'] = df['Salary'].astype(int) # this convert the column datatypes from object into numbers , in this case integer

df.head()
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

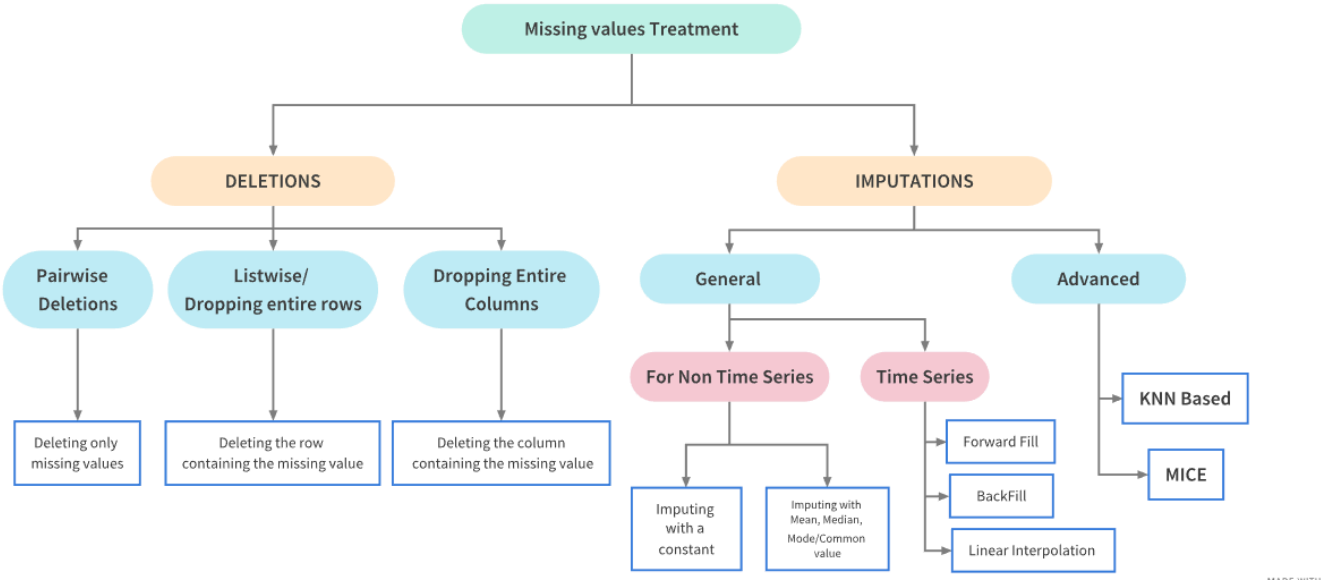
	Timestamp	Age	Industry	Job title	Context	Salary	Compensation	Currency	Other currency	Salary context	Country	State	City	Ove year experi
0	4/27/2021 11:02:10	25-34	Education (Higher Education)	Research and Instruction Librarian	NaN	55000	0.0	USD	NaN	NaN	United States	Massachusetts	Boston	5-7
1	4/27/2021 11:02:22	25-34	Computing or Tech	Change & Internal Communications Manager	NaN	54600	4000.0	GBP	NaN	NaN	United Kingdom	NaN	Cambridge	8 - 10
2	4/27/2021 11:02:38	25-34	Accounting, Banking & Finance	Marketing Specialist	NaN	34000	NaN	USD	NaN	NaN	US	Tennessee	Chattanooga	2 - 4
3	4/27/2021 11:02:41	25-34	Nonprofits	Program Manager	NaN	62000	3000.0	USD	NaN	NaN	USA	Wisconsin	Milwaukee	8 - 10
4	4/27/2021 11:02:42	25-34	Accounting, Banking & Finance	Accounting Manager	NaN	60000	7000.0	USD	NaN	NaN	US	South Carolina	Greenville	8 - 10

```
df.isna().sum()# This shows you the total number of missing values per column
```

```
Timestamp                              0
Age                                    0
Industry                              72
Job title                              0
Context                            20749
Salary                                 0
Compensation                        7270
Currency                               0
Other currency                     27790
Salary context                     24954
Country                                0
State                               4996
City                                  76
Overall years of experience            0
Years of experience in current job     0
Education level                      216
Gender                               167
Race                                 172
dtype: int64
```

1. Incomplete data/Missing values

- [Reference] https://www.kaggle.com/code/parulpandey/a-guide-to-handling-missing-values-in-python#Treating-Missing-values



- There are 3 columns which has a lot of nulls/missing data

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27989 entries, 0 to 27988
Data columns (total 18 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   Timestamp                          27989 non-null  object
 1   Age                                27989 non-null  object
 2   Industry                           27917 non-null  object
 3   Job title                          27989 non-null  object
 4   Context                            7240 non-null   object
 5   Salary                             27989 non-null  int64
 6   Compensation                       20719 non-null  float64
 7   Currency                           27989 non-null  object
 8   Other currency                     199 non-null    object
 9   Salary context                     3035 non-null   object
 10  Country                            27989 non-null  object
 11  State                              22993 non-null  object
 12  City                               27913 non-null  object
 13  Overall years of experience        27989 non-null  object
 14  Years of experience in current job 27989 non-null  object
 15  Education level                    27773 non-null  object
 16  Gender                             27822 non-null  object
 17  Race                               27817 non-null  object
dtypes: float64(1), int64(1), object(16)
memory usage: 3.8+ MB
```

```
#Write a code to return the number of rows with missing values under the column 'Compensation'?
# Hint: use isna().sum()

df['Compensation'].isna().sum()
```

```
7270
```

```
#Write a code that returns the each column name with the number of missing values. This will come in handy later
#Output example as below

df.isna().sum()
```

```
Timestamp                              0
Age                                    0
Industry                              72
Job title                              0
Context                            20749
Salary                                 0
Compensation                        7270
Currency                               0
Other currency                     27790
Salary context                     24954
Country                                0
State                               4996
City                                  76
Overall years of experience            0
Years of experience in current job     0
Education level                      216
Gender                               167
Race                                 172
dtype: int64
```

1.1 Dropping columns with too many missing values

```
#Drop columns 'context', 'Other currency', 'Salary context'

df = df.drop(columns=['Context','Other currency', 'Salary context'])
```

```
df.head()
```

	Timestamp	Age	Industry	Job title	Salary	Compensation	Currency	Country	State	City	Overall years of experience	Years of experience in current job	Education level
0	4/27/2021 11:02:10	25-34	Education (Higher Education)	Research and Instruction Librarian	55000	0.0	USD	United States	Massachusetts	Boston	5-7 years	5-7 years	Master's degree
1	4/27/2021 11:02:22	25-34	Computing or Tech	Change & Internal Communications Manager	54600	4000.0	GBP	United Kingdom	NaN	Cambridge	8 - 10 years	5-7 years	College degree
2	4/27/2021 11:02:38	25-34	Accounting, Banking & Finance	Marketing Specialist	34000	NaN	USD	US	Tennessee	Chattanooga	2 - 4 years	2 - 4 years	College degree
3	4/27/2021 11:02:41	25-34	Nonprofits	Program Manager	62000	3000.0	USD	USA	Wisconsin	Milwaukee	8 - 10 years	5-7 years	College degree
			Accounting,										

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27989 entries, 0 to 27988
Data columns (total 15 columns):
 #   Column                            Non-Null Count  Dtype
---  ------                            --------------  -----
 0   Timestamp                         27989 non-null  object
 1   Age                               27989 non-null  object
 2   Industry                          27917 non-null  object
 3   Job title                         27989 non-null  object
 4   Salary                            27989 non-null  int64
 5   Compensation                      20719 non-null  float64
 6   Currency                          27989 non-null  object
 7   Country                           27989 non-null  object
 8   State                             22993 non-null  object
 9   City                              27913 non-null  object
 10  Overall years of experience       27989 non-null  object
 11  Years of experience in current job 27989 non-null  object
 12  Education level                   27773 non-null  object
 13  Gender                            27822 non-null  object
 14  Race                              27817 non-null  object
dtypes: float64(1), int64(1), object(13)
memory usage: 3.2+ MB
```

1.2 Quick analysis of missing values- Compensation.

- The amount to compensate relates well with the industry

```
x = df[df['Compensation'].isna()]
x.head()
```

	Timestamp	Age	Industry	Job title	Salary	Compensation	Currency	Country	State	City	Overall years of experience	Years of experience in current job	Education level
2	4/27/2021 11:02:38	25-34	Accounting, Banking & Finance	Marketing Specialist	34000	NaN	USD	US	Tennessee	Chattanooga	2 - 4 years	2 - 4 years	College degree
5	4/27/2021 11:02:46	25-34	Education (Higher Education)	Scholarly Publishing Librarian	62000	NaN	USD	USA	New Hampshire	Hanover	8 - 10 years	2 - 4 years	Master's degree
7	4/27/2021 11:03:00	25-34	Education (Primary/Secondary)	Librarian	50000	NaN	USD	United States	Arizona	Yuma	5-7 years	5-7 years	Master's degree
14	4/27/2021 11:03:11	18-24	Health care	Patient care coordinator	32000	NaN	CAD	Canada	NaN	Remote	1 year or less	1 year or less	College degree

- One sensible way is to impute NaN values for Compensation based on the industry. For example below, we have records where other people in the Librarian industry receives zero compensation.

- So similarly, we can impute 0 in the compensation where the industry is Librarian

```
#Let's take a look at compensation for librarian from the original dataframe
df[df['Industry']=='Librarian']
```

	Timestamp	Age	Industry	Job title	Salary	Compensation	Currency	Country	State	
2450	4/27/2021 11:44:24	35-44	Librarian	Digital Initiatives Archivist, Senior	66703	0.0	USD	United States	Florida	Laud

1.3 Dropping missing values

- We can drop missing values here using dropna() function.
- If we do dropna() without specifying any parameter, it will drop rows that has even 1 missing values
- If we do dropna(thresh=2) it will drop rows that has more than 2 missing values
- If we do dropna(how=all) it will drop only rows where all values are missing
- [Ref] https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.dropna.html

```
df_dropna = df.dropna()
df_dropna.info()
#Compare the number of rows with the original dataset
#16974 rows vs originally 27989 rows
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 16974 entries, 0 to 27986
Data columns (total 15 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   Timestamp                          16974 non-null  object
 1   Age                                16974 non-null  object
 2   Industry                           16974 non-null  object
 3   Job title                          16974 non-null  object
 4   Salary                             16974 non-null  int64
 5   Compensation                       16974 non-null  float64
 6   Currency                           16974 non-null  object
 7   Country                            16974 non-null  object
 8   State                              16974 non-null  object
 9   City                               16974 non-null  object
 10  Overall years of experience        16974 non-null  object
 11  Years of experience in current job 16974 non-null  object
 12  Education level                    16974 non-null  object
 13  Gender                             16974 non-null  object
 14  Race                               16974 non-null  object
dtypes: float64(1), int64(1), object(13)
memory usage: 2.1+ MB
```

1.4 Impute missing values with most frequent

- A simple method where we don't analyse per column.
- We can impute missing values for all related columns using most frequent.
- Imputed values can be the mean, median. This is changed in the 'strategy' parameter.
- There are other simple methods for imputing missing values. Eg fill.na()

sklearn.impute.SimpleImputer

class **sklearn.impute.SimpleImputer**(*, *missing_values=nan, strategy='mean', fill_value=None, copy=True, add_indicator=False, keep_empty_features=False*) [source]

Univariate imputer for completing missing values with simple strategies.

Replace missing values using a descriptive statistic (e.g. mean, median, or most frequent) along each column, or using a constant value.

Read more in the User Guide.

```
from sklearn.impute import SimpleImputer

#df2 = df.copy() #so that you are not overwriting the original dataframe, #overiding

#df_impute = df.copy() #many times it is wise to copy the dataframe rather than reassign it
most_frequent_imputer = SimpleImputer(strategy='most_frequent') #change strategy here if want to use median etc.
df.iloc[:,:] = most_frequent_imputer.fit_transform(df)
df.head()
```

	Timestamp	Age	Industry	Job title	Salary	Compensation	Currency	Country	State	City	Overall years of experience	Years of experience in current job	Education level
0	4/27/2021 11:02:10	25-34	Education (Higher Education)	Research and Instruction Librarian	55000	0.0	USD	United States	Massachusetts	Boston	5-7 years	5-7 years	Master's degree
1	4/27/2021 11:02:22	25-34	Computing or Tech	Change & Internal Communications Manager	54600	4000.0	GBP	United Kingdom	California	Cambridge	8 - 10 years	5-7 years	College degree
2	4/27/2021 11:02:38	25-34	Accounting, Banking & Finance	Marketing Specialist	34000	0.0	USD	US	Tennessee	Chattanooga	2 - 4 years	2 - 4 years	College degree
3	4/27/2021 11:02:41	25-34	Nonprofits	Program Manager	62000	3000.0	USD	USA	Wisconsin	Milwaukee	8 - 10 years	5-7 years	College degree
			Accounting,										

```
df.isna().sum()
```

```
Timestamp                           0
Age                                 0
Industry                            0
Job title                           0
Salary                              0
Compensation                        0
Currency                            0
Country                             0
State                               0
City                                0
Overall years of experience         0
Years of experience in current job  0
Education level                     0
Gender                              0
Race                                0
dtype: int64
```

2. Noisy data

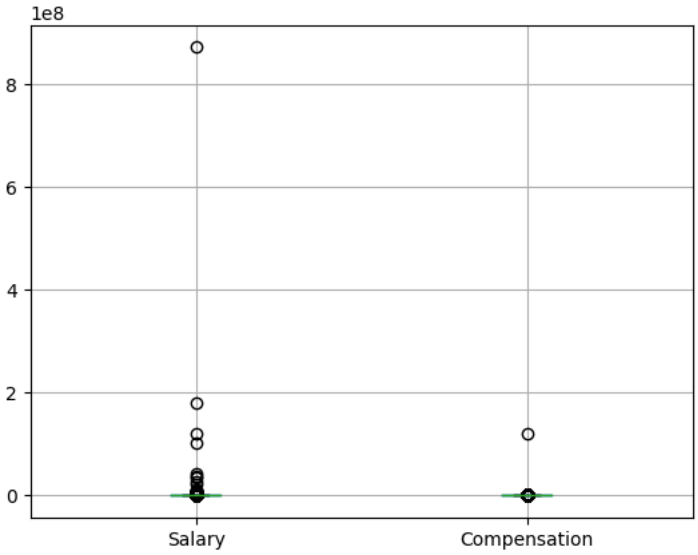
- Noise can be random errors and/or outliers.
- It can also be non-sensible data

2.1 Remove outliers

- Use boxplot or histograms to find outliers (df.boxplot() or df.hist())
- Outliers are values that are waaay higher or lower than most of the data. Usually there are not many data points like this. So we can consider this as noise
- It seems salary and compensation has many outliers. Let's remove them

```
df.boxplot() # There are
```

```
<Axes: >
```

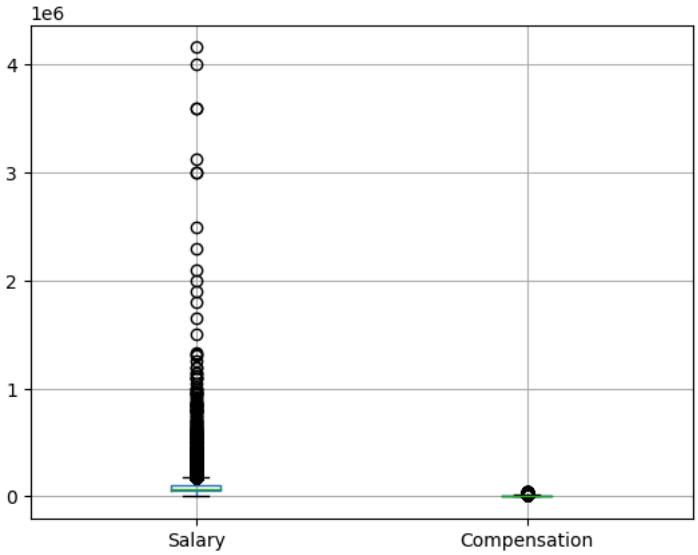


- Can remove using stats [Ref] https://stackoverflow.com/questions/23199796/detect-and-exclude-outliers-in-a-pandas-dataframe
- Can also remove manually

```
from scipy import stats
df=df[(np.abs(stats.zscore(df['Salary'])) < 0.75)] #replace 1.5 with your own threshold
df=df[(np.abs(stats.zscore(df['Compensation'])) < 0.75)]

df.boxplot()
```

```
<Axes: >
```



2.2 Remove non sensible values

- Use df.describe()

```
print(df['Salary'].describe())
print('is it really true that the min salary is zero?::',df['Salary'].min())

'''


salary_zero = df[df['Salary']==0]
print(len(salary_zero))


df[df['Salary']==0].count()
df[df['Salary']==0]['Salary'].count()

df[df['Salary']==0]


'''
```

```
count    2.656900e+04
mean     8.676914e+04
std      8.977519e+04
min      0.000000e+00
25%      5.300000e+04
50%      7.400000e+04
75%      1.030000e+05
max      4.160000e+06
Name: Salary, dtype: float64
is it really true that the min salary is zero?:: 0
'\n\n\nsalary_zero = df[df['Salary']==0]\nprint(len(salary_zero))\n\n\ndf[df['Salary']==0].count()\ndf[df['Salary']==0]['Salary'].count()\n\ndf[d
f['Salary']==0]\n\n\n'
```

```
df[df['Salary']<500]

df = df[df['Salary']>=17000] #any data that has annual salary of less than 17k is non sensible. so we 'drop' it by subset
```

3. Inconsistent data

- Find 'duplicated' rows. For example 'University Malaya' , 'UM', 'Universiti Malaya' should be the same. One way is to use fuzzy matching (fuzzywuzzy package)
- Find typos
- Warning , this is very tedious
- [Ref] https://www.kaggle.com/code/ramjan135/data-cleaning-challenge-inconsistent-data-entry

3.1 Check for each columns. Example is 'Gender'

- Print out the unique items for each columns using df['column].unique()
- You'll see some are not that unique. Using the 'Gender' column, I found that there is 'Other or prefer not to answer' and 'Prefer not to answer'. Let's just group them into 'Other'
- Try checking columns with the lowest number of unique itmes first

```
print('--Columns and their number of unique items--')

df.nunique()
```

```
--Columns and their number of unique items--
Timestamp                           23083
Age                                     7
Industry                             1166
Job title                           13671
Salary                               3410
Compensation                          613
Currency                               11
Country                               342
State                                 125
City                                 4647
Overall years of experience             8
Years of experience in current job      8
Education level                         6
Gender                                  5
Race                                   48
dtype: int64
```

- Check the unique items for Gender

```
#Forcing colab to print out all in the y (stored as ndarray)
import sys
import numpy
numpy.set_printoptions(threshold=sys.maxsize)

#y = df['Industry'].unique()
y = df['Gender'].unique()
print(f'There are {len(y)} unique items ')
print(y)
```

```
There are 5 unique items
['Woman' 'Non-binary' 'Man' 'Other or prefer not to answer'
 'Prefer not to answer']
```

- Remove redundancy by using df.replace()

```
df = df.replace('Other or prefer not to answer','Prefer Not Answer')
#df = df.replace('Prefer not to answer','Prefer Not to Answer')
```

- Check whether the values have been replaced or not

```
y = df['Gender'].unique()
print(f'There are {len(y)} unique items ')
print(y)
```

```
There are 5 unique items
['Woman' 'Non-binary' 'Man' 'Prefer Not Answer' 'Prefer not to answer']
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 26346 entries, 0 to 27988
Data columns (total 15 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   Timestamp                          26346 non-null  object
 1   Age                                26346 non-null  object
 2   Industry                           26346 non-null  object
 3   Job title                          26346 non-null  object
 4   Salary                             26346 non-null  int64
 5   Compensation                       26346 non-null  float64
 6   Currency                           26346 non-null  object
 7   Country                            26346 non-null  object
 8   State                              26346 non-null  object
 9   City                               26346 non-null  object
 10  Overall years of experience        26346 non-null  object
 11  Years of experience in current job 26346 non-null  object
 12  Education level                    26346 non-null  object
 13  Gender                             26346 non-null  object
 14  Race                               26346 non-null  object
dtypes: float64(1), int64(1), object(13)
memory usage: 3.2+ MB
```